

HDOC – Day 25

C Programming Assignment

Name - Arushi Gusain

Sap id - 590033489

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Decreasing Patterns using switch statement, nested loops, arrays and parameterless void functions

Problem requirement: Display a menu and use a switch statement to print the decreasing pattern selected by the user. Each pattern is implemented in a separate parameterless function with no return value.

1. Decreasing right-angled triangle

Input the number of rows n and print a decreasing right-angled triangle using asterisks.

Algorithm

1. START
2. Input n.
3. If $n \leq 0$, display an error and return.
4. For i from n down to 1:
5. Print '*' exactly i times using an inner loop.
6. Move to the next line.
7. STOP

Test Case Table

Test Case	Input	Expected output
1	4	* * * * * * * * * *
2	1	*
3	0	Invalid number of rows

2. Decreasing natural-number pattern

In the first row print natural numbers from 1 to n. Reduce the number of values printed by one in each successive row.

Algorithm

1. START
2. Input n.
3. If $n \leq 0$, display an error and return.
4. For i from n down to 1:
5. For j from 1 to i, print j.
6. Move to the next line.
7. STOP

Test Case Table

Test Case	Input n	Expected Output
1	4	1 2 3 4 1 2 3 1 2 1
2	2	1 2 1
3	-2	Invalid number of rows.

3. Decreasing repeated-number pattern

Begin with n in the first row and decrease the value by one in each successive row. Print each value as many times as its value.

Algorithm

1. START
2. Input n.
3. If $n \leq 0$, display an error and return.
4. For i from n down to 1:
5. Print the value i exactly i times.
6. Move to the next line.
7. STOP

Test Case Table

Test Case	Input n	Expected output
1	4	4 4 4 4 3 3 3 2 2 1
2	3	3 3 3 2 2 1
3	0	Invalid number of rows.

4. Reverse-number triangle

In each row, print natural numbers in descending order, beginning with the number of elements in that row and ending with 1.

Algorithm

1. START
2. Input n.
3. If $n \leq 0$, display an error and return.
4. For i from n down to 1:
5. For j from i down to 1, print j.
6. Move to the next line.
7. STOP

Test Case Table

Test Case	Input n	Expected output
1	4	4 3 2 1 3 2 1 2 1 1
2	2	2 1 1
3	-1	Invalid number of rows

Complete C Program

```
#include <stdio.h>
void decreasingStarTriangle(void);
void decreasingNaturalPattern(void);
void decreasingRepeatedPattern(void);
void reverseNumberTriangle(void);
int main(void)
{
    int choice;
    printf("DECREASING PATTERNS MENU\n");
    printf("1. Decreasing right-angled triangle\n");
    printf("2. Decreasing natural-number pattern\n");
    printf("3. Decreasing repeated-number pattern\n");
    printf("4. Reverse-number triangle\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch (choice)
    {
        case 1:
            decreasingStarTriangle();
            break;
        case 2:
            decreasingNaturalPattern();
            break;
        case 3:
            decreasingRepeatedPattern();
            break;
        case 4:
            reverseNumberTriangle();
            break;
        default:
            printf("Invalid choice.\n");
    }
    return 0;
}
```

```

void decreasingStarTriangle(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    if (n <= 0) {
        printf("Invalid number of rows.\n");
        return;
    }
    for (i = n; i >= 1; i--) {
        for (j = 1; j <= i; j++)
            printf("* ");
        printf("\n");
    }
}

void decreasingNaturalPattern(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    if (n <= 0) {
        printf("Invalid number of rows.\n");
        return;
    }
    for (i = n; i >= 1; i--) {
        for (j = 1; j <= i; j++)
            printf("%d ", j);
        printf("\n");
    }
}

void decreasingRepeatedPattern(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

```

```
if (n <= 0) {
printf("Invalid number of rows.\n");
return;
}
for (i = n; i >= 1; i--) {
for (j = 1; j <= i; j++)
printf("%d ", i);
printf("\n");
}
}

void reverseNumberTriangle(void)
{
int n, i, j;
printf("Enter number of rows: ");
scanf("%d", &n);
if (n <= 0) {
printf("Invalid number of rows.\n");
return;
}
for (i = n; i >= 1; i--) {
for (j = i; j >= 1; j--)
printf("%d ", j);
printf("\n");
}
}
```

Compilation , Execution & Testing

```
(kali㉿kali)-[~]
$ cc hdoc25.c

(kali㉿kali)-[~]
$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 1
Enter number of rows: 4
* * * *
* * *
* *
*
*

(kali㉿kali)-[~]
$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 1
Enter number of rows: 1
*

(kali㉿kali)-[~]
$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 1
Enter number of rows: 0
Invalid number of rows.
```

```
(kali㉿kali)-[~]
$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 2
Enter number of rows: 4
1 2 3 4
1 2 3
1 2
1

(kali㉿kali)-[~]
$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 2
Enter number of rows: 2
1 2
1

(kali㉿kali)-[~]
$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 2
Enter number of rows: -2
Invalid number of rows.
```

```
(kali㉿kali)-[~]
└─$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 3
Enter number of rows: 4
4 4 4 4
3 3 3
2 2
1
```

```
(kali㉿kali)-[~]
└─$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 3
Enter number of rows: 3
3 3 3
2 2
1
```

```
(kali㉿kali)-[~]
└─$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 3
Enter number of rows: 0
Invalid number of rows.
```

```
(kali㉿kali)-[~]
└─$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 4
Enter number of rows: 4
4 3 2 1
3 2 1
2 1
1
```

```
(kali㉿kali)-[~]
└─$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 4
Enter number of rows: 2
2 1
1
```

```
(kali㉿kali)-[~]
└─$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 4
Enter number of rows: -1
Invalid number of rows.
```

